PYTHON CORE & DATA STRUCTURES — 100 MCQ TEST

Comprehensive Assessment covering Data Types, Built-in Functions, Control Flow & Features

Subject:	Python Programming	**Total Questions:**	100 MCQs	**Time Allowed:**	90 Minutes
Topics:	Variables, Data Types (Set, Dict, Frozenset, List, Tuple), Built-in Functions (id, type, input), If-Else, Match-Case, Loops, Python Features				

Section 1: Python Features, Variables & Built-in Functions (Q1 – Q20)

1. Who created the Python programming language?

A) James Gosling
B) Guido van Rossum
C) Dennis Ritchie
D) Bjarne Stroustrup

2. Which of the following is NOT a core feature of Python?

A) Dynamically Typed
B) Interpreted Language
C) Compulsory Manual Memory Management
D) Multi-paradigm Support

3. What function is used to obtain the memory address of an object in Python?

A) `address()`
B) `mem()`
C) `id()`
D) `location()`

4. What is the return type of the built-in `input()` function in Python 3?

A) `int`
B) `str`
C) `float`
D) Dynamic depending on input value

5. Which of the following is a valid variable name in Python?

A) `2my_var`
B) `my-var`
C) `_my_var`
D) `import`

6. What does the `type()` function return?

A) Memory size of the object
B) Data type class of the object
C) Name of the script
D) Variable reference count

7. What is the output of `print(type(1 / 2))` in Python 3?

A) `<class 'int'>`
B) `<class 'float'>`
C) `<class 'double'>`
D) SyntaxError

8. Which keyword is used to declare a variable in Python?

A) `var`
B) `let`
C) `val`
D) No keyword is required

9. Which operator is used for integer floor division in Python?

A) `/`
B) `//`
C) `%`
D) `**`

10. What will be the output of `bool("False")`?

A) `False`
B) `True`
C) `None`
D) ValueError

11. Python uses which mechanism for memory management?

A) Automatic Reference Counting & Garbage Collection
B) Manual free() calls
C) Pointers only
D) Stack allocations only

12. What is the maximum possible length of an identifier in Python?

A) 31 characters
B) 63 characters
C) 79 characters
D) Unspecified (unlimited length)

13. Which statement about Python namespaces is correct?

A) Namespaces are implemented as lists
B) A namespace is a mapping from names to objects
C) Python has only one global namespace
D) Namespaces cannot be nested

14. What will `print(id(a) == id(b))` output if `a = 256` and `b = 256` in standard CPython?

A) `True` (due to integer caching)
B) `False`
C) SyntaxError
D) RuntimeError

15. What will `type([])` evaluate to?

A) `<class 'array'>`
B) `<class 'list'>`
C) `<class 'sequence'>`
D) `<class 'tuple'>`

16. Which of the following is an immutable built-in data type in Python?

A) `list`
B) `dict`
C) `set`
D) `tuple`

17. How can you take multiple space-separated inputs as integers in Python?

A) `input().split()`
B) `map(int, input().split())`
C) `int(input().split())`
D) `input(int).split()`

18. What is the output of `print(0.1 + 0.2 == 0.3)` in Python?

A) `True`
B) `False`
C) `None`
D) TypeError

19. Which function yields the string representation of an object meant for debugging?

A) `str()`
B) `repr()`
C) `print()`
D) `format()`

20. What design philosophy acronym is associated with Python's approach to variable typing?

A) Static Typing
B) Duck Typing
C) Strict Typing
D) Nominal Typing

Section 2: Control Flow — If-Else & Match-Case (Q21 – Q35)

21. Which keyword is used for conditional checks following an `if` in Python?

A) `else if`
B) `elseif`
C) `elif`
D) `then`

22. In Python 3.10+, structural pattern matching is introduced using which keywords?

A) `switch / case`
B) `select / case`
C) `match / case`
D) `do / match`

23. In a `match-case` statement, which symbol acts as a wildcard default case?

A) `*`
B) `_` (underscore)
C) `default`
D) `else`

24. What will be the output of `x = 10; print("A") if x > 15 else print("B")`?

A) A
B) B
C) A B
D) SyntaxError

25. What happens if no `case` matches in a `match` statement and no wildcard case is provided?

A) Throws MatchError
B) Throws ValueError
C) Execution simply continues past the match block
D) Raises SyntaxError

26. Which operator combines multiple conditions such that both must be True?

A) `&&`
B) `and`
C) `&`
D) `both`

27. What is the value of `res` in: `res = "Even" if 4 % 2 == 0 else "Odd"`?

A) "Even"
B) "Odd"
C) True
D) False

28. In `match-case`, how can you combine multiple patterns in a single case?

A) Using comma `case 1, 2:`
B) Using pipe `case 1 | 2:`
C) Using `or` keyword `case 1 or 2:`
D) Using `case [1, 2]:`

29. Can guards (`if` conditions) be added to `case` patterns in Python?

A) No, match-case doesn't support guards
B) Yes, e.g., `case x if x > 0:`
C) Yes, but only with `and` operator
D) Only in default cases

30. What evaluates to True in Python boolean context?

A) `[]` (empty list)
B) `0`
C) `"0"` (non-empty string)
D) `None`

31. What is short-circuit evaluation in Python logical operators?

A) Evaluating expressions from right to left
B) Stopping evaluation as soon as outcome is determined
C) Converting all operands to boolean integers
D) Raising error on undefined variables

32. What is the output of `5 and 10` in Python?

A) `True`
B) `5`
C) `10`
D) `False`

33. What is the output of `0 or "Python"`?

A) `0`
B) `"Python"`
C) `True`
D) `False`

34. Which statement correctly describes indentation in Python control blocks?

A) Indentation is optional
B) Curley braces override indentation
C) Indentation defines code block scope
D) Only tabs are allowed for indentation

35. What is the purpose of the `pass` statement in conditional blocks?

A) Exit the program
B) Skip the current loop iteration
C) Act as a null placeholder statement
D) Return None from the block

Section 3: Loops & Iteration Concepts (Q36 – Q50)

36. What sequence does `range(2, 8, 2)` generate?

A) 2, 4, 6, 8
B) 2, 4, 6
C) 2, 3, 4, 5, 6, 7
D) 2, 5, 8

37. When does the `else` clause of a `for` loop execute in Python?

A) When the loop is terminated by a `break`
B) When the loop completes normally without hitting `break`
C) On every iteration
D) Only if the iterable is empty

38. Which keyword is used to skip the rest of the code inside a loop for the current iteration?

A) `pass`
B) `break`
C) `continue`
D) `skip`

39. What will `for x in range(0): print(x)` output?

A) 0
B) None
C) Nothing (loop body does not run)
D) ValueError

40. How to iterate over both index and value of a list `items` simultaneously?

A) `for i, v in zip(items):`
B) `for i, v in enumerate(items):`
C) `for i, v in items.items():`
D) `for i, v in index(items):`

41. What is the output of: `i = 1; while i < 4: print(i, end=""); i += 1` ?

A) 1234
B) 123
C) 12
D) Infinite loop

42. Which function can pair elements from two equal-length lists into tuples during iteration?

A) `combine()`
B) `pair()`
C) `zip()`
D) `merge()`

43. What happens if a `while` condition never becomes False and contains no `break` ?

A) Python auto-terminates after 1000 cycles
B) Infinite loop (runs until interrupted)
C) RecursionError
D) SyntaxError

44. What does `range(5, 1, -1)` produce?

A) 5, 4, 3, 2, 1
B) 5, 4, 3, 2
C) 4, 3, 2, 1
D) Empty range

45. What is the output of `sum(range(1, 5))` ?

A) 15
B) 10
C) 5
D) 14

46. Which of the following data structures can be directly iterated with a `for` loop?

A) List and Tuple
B) Dictionary and Set
C) String and Generator
D) All of the above

47. What will `[x*2 for x in range(3)]` generate?

A) `[0, 2, 4]`
B) `[2, 4, 6]`
C) `(0, 2, 4)`
D) `[0, 1, 2]`

48. What is the effect of `break` statement inside a nested loop?

A) Terminates all outer loops
B) Terminates only the innermost enclosing loop
C) Skips to next iteration of outer loop
D) Stops entire Python process

49. What is the type of object returned by `range(5)` in Python 3?

A) `list`
B) `tuple`
C) `range` object (iterator/sequence)
D) `generator`

50. In a `while-else` loop structure, if `break` executes, does `else` run?

A) Yes
B) No
C) Only if condition was initially True
D) Depends on Python version

Section 4: Sequential Data Types — List & Tuple (Q51 – Q70)

51. Which statement correctly creates a single-element tuple in Python?

A) `t = (1)`
B) `t = (1,)`
C) `t = tuple(1)`
D) `t = [1]`

52. What is the main difference between a list and a tuple in Python?

A) Lists are ordered; tuples are unordered
B) Lists are mutable; tuples are immutable
C) Tuples hold only integers; lists hold anything
D) Lists cannot be nested

53. What is the result of `lst = [1, 2, 3]; lst.append([4, 5])`?

A) `[1, 2, 3, 4, 5]`
B) `[1, 2, 3, [4, 5]]`
C) `[1, 2, 3, 4]`
D) TypeError

54. Which method adds individual elements of an iterable to an existing list?

A) `append()`
B) `add()`
C) `extend()`
D) `insert()`

55. What does `lst.pop()` do by default when given no arguments?

A) Removes and returns the first element
B) Removes and returns the last element
C) Clears the entire list
D) Returns a random element

56. What does `[1, 2] + [3, 4]` produce?

A) `[4, 6]`
B) `[[1, 2], [3, 4]]`
C) `[1, 2, 3, 4]`
D) TypeError

57. What is the output of `a = (1, 2, 3); a[0] = 99`?

A) `(99, 2, 3)`
B) TypeError: 'tuple' object does not support item assignment
C) ValueError
D) `[99, 2, 3]`

58. How do you slice a list `lst` to reverse it?

A) `lst[0:-1]`
B) `lst[::-1]`
C) `lst[1::-1]`
D) `lst.reverse()` (returns new list)

59. What will `lst = [10, 20, 30]; print(lst.index(20))` print?

A) 1
B) 20
C) 2
D) True

60. What is the complexity of accessing an element by index in a Python list?

A) O(n)
B) O(log n)
C) O(1)
D) O(n^2)

61. What happens if you call `lst.remove(x)` when `x` is NOT in `lst`?

A) Returns None
B) Returns False
C) Raises ValueError
D) Raises KeyError

62. Which method sorts a list in-place?

A) `sorted(lst)`
B) `lst.sort()`
C) `lst.order()`
D) `lst.arrange()`

63. Can a tuple be used as a key in a Python dictionary?

A) Always
B) Never
C) Only if all elements inside the tuple are immutable
D) Only if tuple length is 2

64. What is packing and unpacking in Python tuples?

A) Converting tuples to JSON
B) Assigning multiple values to a tuple and extracting them into variables
C) Compressing tuple memory
D) Encrypting tuple data

65. What does `lst.count(x)` return?

A) Total length of list
B) Number of occurrences of element x
C) Index of x
D) True if x exists

66. What will `(1, 2) * 2` yield?

A) `(2, 4)`
B) `(1, 2, 1, 2)`
C) `((1, 2), (1, 2))`
D) TypeError

67. How do you create an empty tuple?

A) `()`
B) `tuple()`
C) Both A and B
D) `tuple([])` only

68. Which list method inserts an element at a specified index?

A) `lst.add(index, val)`
B) `lst.insert(index, val)`
C) `lst.put(index, val)`
D) `lst.push(index, val)`

69. What is the output of `list("cat")`?

A) `["cat"]`
B) `["c", "a", "t"]`
C) `("c", "a", "t")`
D) Error

70. What is a shallow copy vs deep copy in Python lists?

A) Shallow copy duplicates references; deep copy duplicates recursively
B) Shallow copy is faster; deep copy works only on integers
C) Shallow copy is immutable; deep copy is mutable
D) They are identical

Section 5: Key-Value & Set Data Types — Dict, Set & Frozenset (Q71 – Q100)

71. How do you create an empty dictionary in Python?

A) `{}`
B) `dict()`
C) Both A and B
D) `set()`

72. How do you create an empty set in Python?

A) `{}`
B) `set()`
C) `set({})`
D) `[]`

73. What key requirement must dictionary keys satisfy in Python?

A) Must be strings
B) Must be hashable (immutable)
C) Must be integers
D) Must be sorted

74. What happens when accessing a non-existent key via `d[key]` vs `d.get(key)`?

A) `d[key]` raises KeyError; `get()` returns None
B) Both raise KeyError
C) Both return None
D) `get()` creates the key automatically

75. What is a `frozenset` in Python?

A) A set containing only frozen strings
B) An immutable version of a Python set
C) A set with fixed memory capacity
D) A ordered list of sets

76. Can a `frozenset` be used as a key in a dictionary or element of another set?

A) Yes, because it is hashable
B) No, because it is a set type
C) Only as a key, not in a set
D) Only if empty

77. What does `{1, 2, 2, 3}` evaluate to?

A) `{1, 2, 2, 3}`
B) `{1, 2, 3}`
C) `[1, 2, 3]`
D) SyntaxError

78. Which set operator calculates the union of set A and set B?

A) `A & B`
B) `A | B`
C) `A ^ B`
D) `A - B`

79. Which operator calculates the intersection of set A and set B?

A) `A & B`
B) `A | B`
C) `A + B`
D) `A % B`

80. What is the output of `s = {1, 2}; s.add(3); print(s)`?

A) `{1, 2, 3}`
B) `{1, 2, (3)}`
C) `[1, 2, 3]`
D) AttributeError

81. How does `set.discard(x)` differ from `set.remove(x)`?

A) `discard()` raises KeyError if x missing; `remove()` does not
B) `remove()` raises KeyError if x missing; `discard()` does not
C) `discard()` works only on strings
D) They are identical

82. Since Python 3.7+, dictionary insertion order is:

A) Arbitrary / Unordered
B) Guaranteed to be preserved
C) Sorted alphabetically by key
D) Sorted by size of value

83. Which dictionary method returns all key-value pairs as tuple pairs?

A) `d.keys()`
B) `d.values()`
C) `d.items()`
D) `d.pairs()`

84. What is the result of `d = {"a": 1}; d.update({"b": 2}); print(d)` ?

A) `{"a": 1, "b": 2}`
B) `{"a": 1}`
C) `{"b": 2}`
D) TypeError

85. What does `d.popitem()` do in modern Python?

A) Removes and returns an arbitrary key-value pair
B) Removes and returns the last inserted key-value pair
C) Removes the first inserted key-value pair
D) Clears the dictionary

86. What will `s = frozenset([1, 2]); s.add(3)` do?

A) Adds 3 to s
B) Creates a new frozenset
C) Raises AttributeError
D) Converts s to set

87. What is symmetric difference between two sets?

A) Elements in both sets
B) Elements in either set, but NOT in both
C) Elements in first set only
D) All unique elements combined

88. What operator or method represents symmetric difference?

A) `^` / `symmetric_difference()`
B) `&` / `intersection()`
C) `-` / `difference()`
D) `+` / `union()`

89. What is the output of `len({"a": 1, "b": 2, "a": 3})` ?

A) 3
B) 2
C) 1
D) KeyError

90. Can a list be added as an element inside a Python set?

A) Yes
B) No, because lists are unhashable (mutable)
C) Only if list contains integers
D) Yes, if converted automatically

91. Which method clears all items from a dictionary `d` ?

A) `d.removeAll()`
B) `d.clear()`
C) `d.empty()`
D) `del d`

92. What will `print("a" in {"a": 1, "b": 2})` check?

A) Checks if "a" exists among keys
B) Checks if "a" exists among values
C) Checks both keys and values
D) Raises SyntaxError

93. What is dict comprehension syntax?

A) `{k: v for k, v in iterable}`
B) `[k: v for k, v in iterable]`
C) `(k: v for k, v in iterable)`
D) `dict(k->v for k,v in iterable)`

94. How to check if set A is a subset of set B?

A) `A <= B` or `A.issubset(B)`
B) `A in B`
C) `A.contains(B)`
D) `A == B`

95. What operator merges two dictionaries in Python 3.9+?

A) `+`
B) `|`
C) `&`
D) `=>`

96. What is the output of `set("hello")`?

A) `{'h', 'e', 'l', 'l', 'o'}`
B) Set containing unique characters: `{'h', 'e', 'l', 'o'}`
C) `{"hello"}`
D) TypeError

97. What does `d.setdefault("c", 3)` do if key "c" is NOT present in `d`?

A) Sets `d["c"] = 3` and returns 3
B) Returns None without modifying d
C) Raises KeyError
D) Overwrites existing keys

98. Is elements order guaranteed in a standard Python `set`?

A) Yes, insertion order
B) Yes, sorted order
C) No, sets are unordered collections
D) Yes, reverse insertion order

99. Which data type is mutable?

A) `frozenset`
B) `tuple`
C) `dict`
D) `str`

100. What is the primary advantage of sets/dict lookups over list lookups?

A) Lower memory footprint
B) Average O(1) time complexity for membership tests via hashing
C) Retains element order automatically
D) Allows duplicate values

Answer Key (Q1 – Q100)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	21	C	41	B	61	C	81	B
2	C	22	C	42	C	62	B	82	B
3	C	23	B	43	B	63	C	83	C
4	B	24	B	44	B	64	B	84	A
5	C	25	C	45	B	65	B	85	B
6	B	26	B	46	D	66	B	86	C
7	B	27	A	47	A	67	C	87	B
8	D	28	B	48	B	68	B	88	A

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
9	B	29	B	49	C	69	B	89	B
10	B	30	C	50	B	70	A	90	B
11	A	31	B	51	B	71	C	91	B
12	D	32	C	52	B	72	B	92	A
13	B	33	B	53	B	73	B	93	A
14	A	34	C	54	C	74	A	94	A
15	B	35	C	55	B	75	B	95	B
16	D	36	B	56	C	76	A	96	B
17	B	37	B	57	B	77	B	97	A
18	B	38	C	58	B	78	B	98	C
19	B	39	C	59	A	79	A	99	C
20	B	40	B	60	C	80	A	100	B